Multivariate Calculus Lecture Notes
Section 12.4 – Cross Product

I. Introduction
   a. Another way to multiply vectors involves the *cross product*.
   b. Unlike the dot product, the cross product yields a vector.
   c. Determinant form calculation
      i. Let $\mathbf{u} = \langle u_1, u_2, u_3 \rangle$, $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$.
      ii. Then

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = \begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix} \mathbf{k}$$
$$= (u_2 v_3 - v_2 u_3)\mathbf{i} - (u_1 v_3 - v_1 u_3)\mathbf{j} + (u_1 v_2 - v_1 u_2)\mathbf{k}.$$

   d. Note the minus sign in front of the $\mathbf{j}$ component.
   e. The cross product is only defined for three-dimensional vectors, not for 2D vectors or any vectors higher than 3D.

II. Examples: Let $\mathbf{u} = \langle 1, 3, 4 \rangle$, $\mathbf{v} = \langle 2, 7, -5 \rangle$.

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 3 & 4 \\ 2 & 7 & -5 \end{vmatrix} = (-15 - 28)\mathbf{i} - (-5 - 8)\mathbf{j} + (7 - 6)\mathbf{k} = -43\mathbf{i} + 13\mathbf{j} + \mathbf{k}$$

$$\mathbf{v} \times \mathbf{u} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 7 & -5 \\ 1 & 3 & 4 \end{vmatrix} = (28 + 15)\mathbf{i} - (8 + 5)\mathbf{j} + (6 - 7)\mathbf{k} = 43\mathbf{i} - 13\mathbf{j} - \mathbf{k}$$

$$\mathbf{u} \times \mathbf{u} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 3 & 4 \\ 1 & 3 & 4 \end{vmatrix} = (12 - 12)\mathbf{i} - (4 - 4)\mathbf{j} + (3 - 3)\mathbf{k} = \mathbf{0}$$

III. Geometric properties
   a. The vector $\mathbf{u} \times \mathbf{v}$ is orthogonal to both $\mathbf{u}$ and $\mathbf{v}$.
      - For example, $\langle 1, 3, 4 \rangle \cdot \langle -43, 13, 1 \rangle = -43 + 39 + 4 = 0$.
   b. The direction of $\mathbf{u} \times \mathbf{v}$ is such that $\mathbf{u}, \mathbf{v}$, and $\mathbf{u} \times \mathbf{v}$ form a right-handed system.
      - Again, this means that if you curl your fingers through the angle from $\mathbf{u}$ to $\mathbf{v}$ your thumb will be pointing in the direction of $\mathbf{u} \times \mathbf{v}$.
   c. $||\mathbf{u} \times \mathbf{v}|| = ||\mathbf{u}|| \, ||\mathbf{v}|| \sin\theta$, where θ is the angle between $\mathbf{u}$ and $\mathbf{v}$.
      - Compare it with the dot product: $\mathbf{u} \cdot \mathbf{v} = ||\mathbf{u}|| \, ||\mathbf{v}|| \cos\theta$.
   d. $\mathbf{u} \times \mathbf{v} = \mathbf{0}$ if and only if $\mathbf{u}$ and $\mathbf{v}$ are parallel (i.e., scalar multiples of each other).
   e. $||\mathbf{u} \times \mathbf{v}||$ is the area of the parallelogram having $\mathbf{u}$ and $\mathbf{v}$ as adjacent sides.
      i. Graphically:

      ii. Area of parallelogram is base length times height.
      iii. Base length is $||\mathbf{u}||$, and height is $||\mathbf{v}|| \sin\theta$.
      iv. From the geometric definition of the cross product, $||\mathbf{u} \times \mathbf{v}|| = ||\mathbf{u}|| \, ||\mathbf{v}|| \sin\theta$.

IV. Algebraic properties
  a. $\mathbf{u} \times \mathbf{v} = -\mathbf{v} \times \mathbf{u}$
  b. $\mathbf{u} \times (\mathbf{v} + \mathbf{w}) = \mathbf{u} \times \mathbf{v} + \mathbf{u} \times \mathbf{w}$
  c. $\mathbf{u} \times \mathbf{0} = \mathbf{0} = \mathbf{0} \times \mathbf{u}$
  d. $\mathbf{u} \times \mathbf{u} = \mathbf{0}$
  e. $\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) = (\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}$
  f. $\lambda(\mathbf{u} \times \mathbf{v}) = (\lambda\mathbf{u}) \times \mathbf{v} = \mathbf{u} \times (\lambda\mathbf{v})$

V. Conceptual examples with $\mathbf{i}, \mathbf{j}$, and $\mathbf{k}$
  a. $\mathbf{i} \times \mathbf{j}$
    i. The magnitude of both vectors is 1, and the angle between them is $\pi/2$, so $||\mathbf{i} \times \mathbf{j}|| = 1$.
    ii. Using the right-hand rule, the direction of $\mathbf{i} \times \mathbf{j}$ is up, so $\mathbf{i} \times \mathbf{j} = \mathbf{k}$.
    iii. Algebraically, we have

$$\mathbf{i} \times \mathbf{j} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{vmatrix} = 0\mathbf{i} - 0\mathbf{j} + \mathbf{k}.$$

  b. $\mathbf{i} \times \mathbf{k}$
    i. The geometric interpretation gives $||\mathbf{i} \times \mathbf{k}|| = 1$.
    ii. The right-hand rule yields $\mathbf{i} \times \mathbf{k} = -\mathbf{j}$.
  c. $\mathbf{j} \times \mathbf{k}$
    i. The magnitude will again be 1.
    ii. The right-hand rule yields $\mathbf{j} \times \mathbf{k} = \mathbf{i}$.
  d. Find $(\mathbf{i} + \mathbf{j}) \times (\mathbf{i} - \mathbf{j})$, geometrically.
    i. The magnitudes of each are $\sqrt{2}$.
    ii. In the xy-plane, the two vectors look like this:

    iii. So the angle between them will be $\pi/2$.
    iv. This yields $||(\mathbf{i} + \mathbf{j}) \times (\mathbf{i} - \mathbf{j})|| = 2$.
    v. For direction, we use the right-hand rule to get $-2\mathbf{k}$.

VI. Triple scalar product
  a. What is the volume of a parallelpiped with $\mathbf{u}, \mathbf{v}$, and $\mathbf{w}$ as adjacent sides?
    i. Graphically:

    ii. $||\mathbf{v} \times \mathbf{w}||$ is the area of the base of the parallelpiped
    iii. $||\mathbf{u}_{||\mathbf{v}\times\mathbf{w}}||$ is the height of the parallelpiped
    iv. The volume of the parallelpiped is the area of its base times its height ($V = Bh$).

v. Thus we have

$$V = ||\mathbf{v}\times\mathbf{w}||\,||\mathbf{u}_{||\mathbf{v}\times\mathbf{w}}|| = ||\mathbf{v}\times\mathbf{w}||\,\left\|\frac{\mathbf{u}\cdot(\mathbf{v}\times\mathbf{w})}{||\mathbf{v}\times\mathbf{w}||^2}\mathbf{v}\times\mathbf{w}\right\|$$
$$= ||\mathbf{v}\times\mathbf{w}||\,\frac{|\mathbf{u}\cdot(\mathbf{v}\times\mathbf{w})|}{||\mathbf{v}\times\mathbf{w}||^2}\,\|\mathbf{v}\times\mathbf{w}\| = |\mathbf{u}\cdot(\mathbf{v}\times\mathbf{w})|.$$

b. The *triple scalar product* of the vectors $\mathbf{u},\mathbf{v}$, and $\mathbf{w}$ (in that order) is given by $\mathbf{u}\cdot(\mathbf{v}\times\mathbf{w})$, or, equivalently, $(\mathbf{u}\times\mathbf{v})\cdot\mathbf{w}$.

c. In determinant form, this is

$$\mathbf{u}\cdot(\mathbf{v}\times\mathbf{w}) = \begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix} = \det\begin{pmatrix}\mathbf{u}\\ \mathbf{v}\\ \mathbf{w}\end{pmatrix}.$$

VII. Final example: Verify that the points $(1,1,1)$, $(2,3,4)$, $(6,5,2)$, and $(7,7,5)$ are the vertices of a parallelogram, and find the area of the parallelogram.

a. Graphically:

b. First, verify that the points are the vertices of a parallelogram.
   i. $(1,1,1)$ and $(2,3,4)$ produce $\langle 1,2,3\rangle$
   ii. $(1,1,1)$ and $(6,5,2)$ produce $\langle 5,4,1\rangle$
   iii. $(6,5,2)$ and $(7,7,5)$ produce $\langle 1,2,3\rangle$
   iv. $(2,3,4)$ and $(7,7,5)$ produce $\langle 5,4,1\rangle$

c. The area is $||\mathbf{u}\times\mathbf{v}||$, where $\mathbf{u} = \langle 1,2,3\rangle$ and $\mathbf{v} = \langle 5,4,1\rangle$.
   i. The cross product is

$$\mathbf{u}\times\mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2 & 3 \\ 5 & 4 & 1 \end{vmatrix} = (2-12)\mathbf{i} - (1-15)\mathbf{j} + (4-10)\mathbf{k} = -10\mathbf{i} + 14\mathbf{j} - 6\mathbf{k}$$

   ii. And the area is thus $||\mathbf{u}\times\mathbf{v}|| = \sqrt{100+196+36} = \sqrt{332} = 2\sqrt{83}$.

Homework: Section 12.4 #9, 12, 21, 24, 28, 38, 41, 44, 49, 58a.